Unit 1 Guided Notes — The Real Number System

Classify it, root it, estimate it, convert it, and order it.

Standards: NC.8.NS.1 · NC.8.NS.2

Name: _____ **Date:** _____ **Period:** _____

🎯 **By the end of this unit, I can…**

- Sort numbers into the real number system (rational vs. irrational)
- Find perfect squares, perfect cubes, and their roots
- Estimate square roots on a number line and to the nearest tenth
- Convert between fractions and decimals, including repeating decimals
- Compare and order real numbers from least to greatest

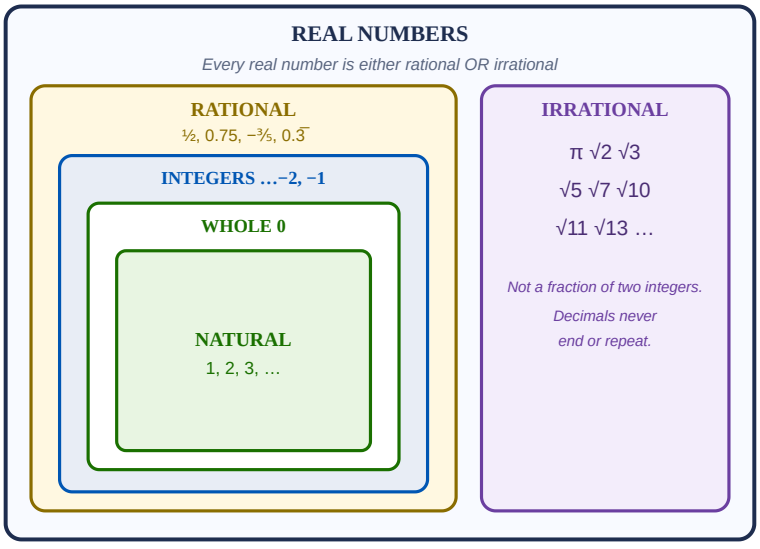
Fill in the blanks and show your work in the space provided. Worked examples are done for you — use them as models.

1 Number Classification

Every real number belongs to one or more of these sets. Write the definition for each:

Set	Examples	Definition
Natural Numbers	1, 2, 3, 4, …	
Whole Numbers	0, 1, 2, 3, …	
Integers	…, −2, −1, 0, 1, 2, …	
Rational Numbers	½, 0.75, −⅗, $0.\overline{3}$	
Irrational Numbers	π, √2, √3	

The Real Number System — Nested Sets



Match the Set to its Definition

Draw a line from each set to the matching definition.

Set	Definition
Natural Numbers	Whole numbers + their negatives
Whole Numbers	Counting numbers: 1, 2, 3, …
Integers	Non-repeating, non-terminating decimal
Irrational Numbers	Naturals + zero

CHECK **Which number is *irrational*? Circle one.**

`0.5̄5` `√49` `√7` `−3/4`

ERROR ANALYSIS **A student claims:** "√49 is irrational because it has a square-root symbol." What is the mistake?

2 Perfect Squares & Square Roots

📌 **THE RELATIONSHIP:** Squaring and square-rooting are INVERSE OPERATIONS — they undo each other. If $5^2 = 25$, then $\sqrt{25} = 5$.

A **perfect square** is a whole number times itself ($n \times n = n^2$). Its **square root** takes you back to that whole number, so √(a perfect square) is always a whole number.

Perfect Squares 1–15 — Reference Chart

n	n^2	n	n^2	n	n^2
1	1	6	36	11	121
2	4	7	49	12	144
3	9	8	64	13	169
4	16	9	81	14	196
5	25	10	100	15	225

💡 **READ IT BOTH WAYS:** $12^2 = 144$ means $\sqrt{144} = 12$. Every row is two facts in one.

CHECK **Circle the perfect square:** **√169 =** _____

30	50	144	200

👉 **NOT A PERFECT SQUARE?** Then the root is irrational (like √7 or √20) — we estimate those in Section 4.

3 Perfect Cubes & Cube Roots

📌 **THE RELATIONSHIP:** Cubing and cube-rooting are INVERSE OPERATIONS too. If $5^3 = 125$, then $\sqrt[3]{125} = 5$. The cube-root symbol carries a small 3: $\sqrt[3]{}$.

A **perfect cube** is a whole number used as a factor three times ($n \times n \times n = n^3$). Its **cube root** undoes that.

Perfect Cubes 1–10 — Reference Chart

n	n^3	n	n^3
1	1	6	216
2	8	7	343
3	27	8	512
4	64	9	729
5	125	10	1000

💡 **SQUARE ROOT vs CUBE ROOT:** √64 asks "what number times *itself* = 64?" → 8. $\sqrt[3]{64}$ asks "what number used *three times* = 64?" → 4. Same number underneath, different answer — watch the little 3!

$\sqrt[3]{27}$ = _____ **$\sqrt[3]{1000}$ =** _____

4 Estimating Square Roots

📌 **THE BIG IDEA:** When a number is NOT a perfect square, its square root is irrational — it never ends or repeats. We can't write it exactly, but we can *estimate* it. Two tools: a number line and a step-by-step algorithm.

Method 1 — Place it on a Number Line

1. Find the two perfect squares your number lands **between**.
2. Take the √ of each — those are two back-to-back whole numbers. Your root lands between them.
3. Decide which one it's **closer** to: is your number nearer the lower or the upper perfect square?
4. Place it on the number line.

Example (model): place √20

16 < **20** < **25**, so √20 is between **4** and **5**. Since 20 is closer to 16, √20 sits a little below the middle — closer to 4.



Method 2 — The Estimating Algorithm (to the nearest tenth)

*Color key: **gold = perfect squares** · **green = the roots** · **blue = your number**.*

Example (model): estimate √45

Step 1 — Trap your number between two perfect squares.

$$36 < 45 < 49$$

√36 = **6** · √49 = **7** → √45 is between **6** and **7**. Smaller root = **6**.

Step 2 — Build the fraction: how far in is your number?

$$\frac{\text{your number} - \text{lower square}}{\text{upper square} - \text{lower square}} = \frac{45 - 36 = 9}{49 - 36 = 13} = 9 \div 13 \approx 0.7$$

The fraction is always squares on top and bottom — never the roots.

Step 3 — Add the fraction to the smaller root.

$$\sqrt{45} \approx 6 + 0.7 = 6.7 \quad \text{(calculator: 6.708…)}$$

💡 **IN SHORT:** √n ≈ (smaller root) + (n − lower square) ÷ (upper square − lower square). This is an *estimate* — it can be off by about a tenth from a calculator, and that's fine.

⚠ **DON'T MIX THEM UP:** the number you **add** is a **root**. The numbers **inside the fraction** are the **perfect squares**. Roots to add, squares to subtract.

YOUR TURN **Estimate √30 to the nearest tenth.** (Hint: 25 < 30 < 36) — show your steps below.

√30 ≈ _____

📌 **NEGATIVE ROOTS — JUST FLIP THE SIGN:** estimate the positive root first, then make it negative. Example: $-\sqrt{40} \rightarrow \sqrt{40} \approx 6.3$, so $-\sqrt{40} \approx -6.3$.

🧭 **WHICH METHOD?** Use the number line for "between which whole numbers." Use the algorithm for a decimal to the nearest tenth.

5 Converting Fractions ↔ Decimals

Fraction → Decimal

Divide the numerator by the denominator (top ÷ bottom).

Fraction	Division	Decimal
3/4	3 ÷ 4	
1/8	1 ÷ 8	
1/3	1 ÷ 3	

💡 **KEY INSIGHT:** Repeating decimals are _____ because they can be written as fractions. Non-repeating, non-terminating decimals (like π) are _____. Use a **bar** over the digits that repeat: $0.\overline{3}$ means 0.3333…

Decimal → Fraction (terminating)

Decimal	Read / write	Fraction (simplest form)
0.75	75 hundredths → 75/100	
0.6	6 tenths → 6/10	
0.125	125 thousandths → 125/1000	

Repeating Decimal → Fraction: The 9's Rule

📌 **THE 9's RULE:** the repeating digits go in the NUMERATOR. The DENOMINATOR is made of 9's — one 9 for each repeating digit.

Repeating digits	Denominator	Example	Simplified
1 digit		$0.\overline{3} = 3/9$	
2 digits		$0.\overline{27} = 27/99$	
3 digits		$0.\overline{125} = 125/999$	

Worked Example (model): convert $0.\overline{27}$ (0.272727…) to a fraction

Step 1 — Count the repeating digits: 2 digits repeat (the block "27").

Step 2 — Denominator = that many 9's: 2 digits → 99.

Step 3 — Numerator = the repeating block: 27/99.

Step 4 — Simplify: 27/99 = **3/11**.

⚠ **IMPORTANT:** the 9's rule only works when the decimal starts repeating *right after* the decimal point. In 8th grade, we only use decimals like that, so the 9's rule always works. Always simplify!

CHECK **Which decimal equals 3/8 ? Circle one.**

`0.38` `0.375` `0.83` `0.625`

6 Ordering Numbers Least to Greatest

📌 **THE STRATEGY:** to compare a mix of fractions, decimals, mixed numbers, and roots — turn *everything* into a decimal first, then order.

Guided Example — Order √5, 2¼, 7/3, 2.1

Number	Decimal	How?
√5		Estimating algorithm (between 4 and 9)
2¼		2 + (1 ÷ 4)
7/3		7 ÷ 3
2.1		Already a decimal

Order (least → greatest): _____ < _____ < _____ < _____

Plot each on the number line, then read them left to right:



💡 **MIXED NUMBERS:** turn them into a decimal — whole number + (numerator ÷ denominator). Example: 3⅖ = 3 + (2 ÷ 5) = 3.4.

⚠ **WATCH OUT:** line up the same number of decimal places before comparing, and estimate any irrational roots *first*.

ERROR ANALYSIS To order 1.5, √2, 7/4, √3 a student wrote 1.5 < √2 < √3 < 7/4. Is this right? Explain and fix it.

✏ Part 2 — Practice Problems

Show all work. Circle your final answer.

A Classify Each Number

Rational = can be written as a fraction (terminating or repeating decimal). Irrational = non-repeating, non-terminating.

1. 0.333… _____
2. π _____
3. √49 _____
4. √10 _____
5. −3/8 _____
6. 0.1212… _____
7. √144 _____
8. 0.101001… _____
9. 2/7 _____
10. √2 _____
11. −√81 _____
12. 3.14 _____

B Estimate Each Square Root (nearest tenth)

Estimate each root to the nearest tenth. For 1–6, use the number line: write the two whole numbers your root falls between in the boxes, mark the point, then read the estimate. For 7–11, use the algorithm and show your work. A leading − just flips the sign.

1. √5 ≈ _____



2. √10 ≈ _____



3. √20 ≈ _____



4. √30 ≈ _____



5. $\sqrt{50} \approx$ _____



6. $\sqrt{8} \approx$ _____



7. $\sqrt{15} \approx$ _____

8. $\sqrt{40} \approx$ _____

9. $\sqrt{65} \approx$ _____

10. $\sqrt{90} \approx$ _____

11. $-\sqrt{50} \approx$ _____

C Order from Least to Greatest

Convert each number to a decimal, then arrange from least to greatest.

1. Order: π, √10, 10/3, 3.5

Write each as a decimal, then order them:

Least → greatest: _____

2. Order: 1.5, √2, 7/4, √3

Write each as a decimal, then order them:

Least → greatest: _____

3. Order: √5, 2.5, 11/5, √7

Write each as a decimal, then order them:

Least → greatest: _____

4. Order: 4.2, √18, 13/3, √20

Write each as a decimal, then order them:

Least → greatest: _____

5. Order: −2.5, −√5, −8/3

Write each as a decimal, then order them:

Least → greatest: _____

6. Order: √50, 7.1, 22/3, √52

Write each as a decimal, then order them:

Least → greatest: _____

7. Order: 5/6, √0.5, 0.9, 7/8

Write each as a decimal, then order them:

Least → greatest: _____

8. Order: √30, 5.5, 17/3, √32

Write each as a decimal, then order them:

Least → greatest: _____